

Wild Atlantic Way Rugby Club Database

DATABASE DESIGN AND DEVELOPMENT

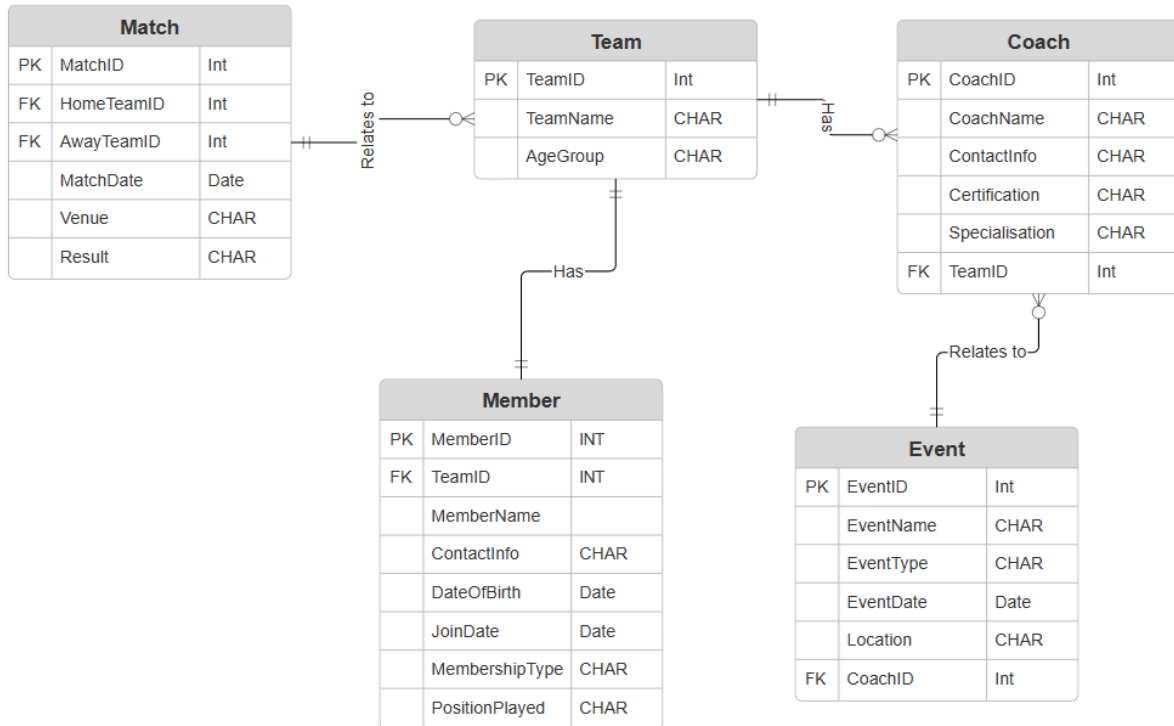
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Part 1. - ERD Diagram

Figure 1.1 ERD Diagram Wild Atlantic Way Rugby Club



In order to create the tables due to reserved words, I have changed the Match Entity name to GameMatch, the Event entity name to TeamEvent and the Member Entity name to ClubMember.

Part 2 – Creation Of The Database

Part 2.1 Creation of The Entities

To create the Database I have assigned the name WildAtlanticWayRugbyClub To the Schema.

I then created the above entities using the following SQL on the DBMS MySQL WorkBench.

```
Create Table Team (  
TeamID Int Not Null,  
TeamName CHAR(50),  
AgeGroup Char(50),  
Primary Key (TeamID)  
);
```

```
Create Table GameMatch (  
MatchID Int Not Null,  
HomeTeamID Int Not Null,  
AwayTeamID Int Not Null,  
MatchDate date,  
Venue CHAR(50),  
Result CHAR(50),  
Primary key (MatchID),  
Foreign key (HomeTeamID) References Team (TeamID),  
Foreign key (AwayTeamID) References Team (TeamID)  
);
```

```
Create Table Coach (  
CoachID Int Not Null,  
CoachName Char(50),  
ContactInfo CHAR(30),  
Certification CHAR(100),  
Specialisation Char(100),  
TeamID int Not Null,  
Primary key (CoachID),  
Foreign Key (TeamID) References Team (TeamID)  
);
```

```
Create Table TeamEvent (  
EventID Int Not Null,  
EventName CHAR(100),  
EventType Char(100),  
EventDate date,  
Location Char(100),  
CoachID Int Not Null,  
Primary key (EventID),  
Foreign Key (CoachID) References Coach (CoachID)  
);
```

```
Create Table ClubMember (  
MemberID Int Not Null,  
TeamID Int Not Null,  
MemberName CHAR(50),  
ContactInfo Char(30),  
DateOfBirth Date,  
JoinDate date,  
MembershipType Char(50),  
PositionPlayed Char(50),  
Primary key (MemberID),  
Foreign key (TeamID) References Team (TeamID)  
);
```

Part 2.1 – Creation Of Users

For The Database I Have created 3 Users Chairperson, Secretary and Membership Secretary.

I have done this using the following Code

```
Create User Chairperson@localhost identified by 'Chair1';
```

```
Create User Secretary@localhost identified by 'Admin1';
```

```
Create User Membership@localhost identified by 'Members1';
```

I have granted all privileges to the chairperson of the club using the following command

```
Grant All Privileges on *.* To Chairperson@localhost;
```

I have then granted the relevant database privileges to the other 2 users, as well as revoking a privilege that I realised wouldn't be needed.

```
Grant Insert, Select, Update, Delete on clubmember To Secretary@localhost;
```

```
Grant Insert, Select, Update, Delete on coach.* To Secretary@localhost;
```

```
Grant Insert, Select, Update, Delete on GameMatch.* To Secretary@localhost;
```

```
Grant Insert, Select, Update, Delete on Team.* To Secretary@localhost;
```

```
Grant Insert, Select, Update, Delete on TeamEvent.* To Secretary@localhost;
```

```
Grant Insert, Select, Update, Delete on clubmember.* To Membership@localhost;
```

```
Grant Insert, Select, Update, Delete on Coach.* To Membership@localhost;
```

```
Grant Insert, Select, Update, Delete on GameMatch.* To Membership@localhost;
```

Revoke Insert, Select, Update, Delete on GameMatch.* from Membership@localhost;

Part 2.2 – Populating Tables

I have added Test Data to all the tables using the following commands.

Insert into Team (

TeamID, TeamName, AgeGroup)

values

(1, 'WAW Mens First Team', 'Senior'),
(2, 'WAW Mens Second Team', 'Senior'),
(3, 'U18s Boys Team', '16-18'),
(4, 'U16s Boys Team', '14-16'),
(5, 'Non Club Team', Null);

Insert into Coach (

CoachID, CoachName, ContactInfo, Certification, Specialisation, TeamID)

values

(1, 'Tom Brady', '0876594324', 'Level 3', 'Head Coach', 1),
(2, 'Mark Vaughan', '0897658767', 'Level 3', 'Forwards Coach', 1),
(3, 'Paul Squire', '0876567908', 'Level 3', 'Backs Coach', 1),
(4, 'Kieran Connolly', '0897676543', 'Level 3', 'Youth Development', 3),
(5, 'John OConnor', '0896754311', 'Level 3', 'Forwards Coach', 2),
(6, 'Margaret OGorman', '0854345676', 'Level 2', 'Children Development', 4);

Insert into ClubMember (

MemberID, TeamID, MemberName, ContactInfo, DateOfBirth, JoinDate,
MembershipType, PositionPlayed)

Values

(1, 1, 'Craig Scott', '0879897654', '1989-08-09', '2000-08-31', 'player', 'Scrumhalf'),
(2, 2, 'Robert Brown', '0876565432', '1984-04-28', '2002-09-08', 'player', 'Prop'),
(3, 4, 'Mark Langley', '0858786745', '2009-08-04', '2019-08-25', 'player', 'Hooker');

Insert into TeamEvent(

EventID, EventName, EventType, EventDate, Location, CoachID)

values

(1, 'CampOut', 'Fundraiser', '2024-08-20', 'Rugby Club', 6),
(2, 'Training Camp', 'Training', '2024-04-05', 'Santa Ponza', 1),

```
(3, 'Club Barbecue and Awards', 'Social', '2024-07-31', 'ClubHouse', 2);
```

```
Insert into gameMatch (  
MatchID, HomeTeamID, AwayTeamID, MatchDate, Venue, Result)  
values  
(1, 1, 5, '2023-10-27', 'WAW ClubHouse', '24:12 WAW Win'),  
(2, 5, 2, '2023-10-27', 'Limerick RFC', '28:07 WAW Lose');
```

Part 3 – Demonstrating use of queries

Part 3.1 – Accessing the data

Using Codd's Law Number 1, I can demonstrate through the use of primary keys, I can access and retrieve data based on the primary key CoachID

```
Select * from Coach Where CoachID = 3;  
Select * from TeamEvent Where EventID = 2;
```

Using an Inner Join based on the TeamID Primary Key in the teams Table, I can join the tables coach and team using the following code, to show the name of which coach coaches which team. This demonstrates Codd's rule number 10, as if we did not have the primary key referencing the foreign key we would not be able to join the tables, and yet the data is still independent as no data has been changed on the individual tables during the join process.

```
Select Team.TeamName, Team.AgeGroup, Coach.CoachName  
From team  
Inner Join Coach On Team.teamID = Coach.TeamID;
```

Views can also be created for Things like coaches, needing to know what players are around to play what positions.

I have demonstrated this with the following code

```
Create View MembershipTeam as  
Select MemberName, PositionPlayed, TeamID  
From ClubMember
```

Where MembershipType = 'Player';

Select * From MembershipTeam;

One of the events needed to be cancelled, so using the delete feature the club barbecue was removed from the database,

Delete from TeamEvent Where EventID = 3;

Codd's Rule 7 has also been demonstrated during the population of data in the database.

An example below, shows multiple rows of data inserted using one statement

Insert into Team (

TeamID, TeamName, AgeGroup)

values

(1, 'WAW Mens First Team', 'Senior'),

(2, 'WAW Mens Second Team', 'Senior'),

(3, 'U18s Boys Team', '16-18'),

(4, 'U16s Boys Team', '14-16'),

(5, 'Non Club Team', Null);